Building a unified science of sensorimotor control with Densely Observable Measurment Environments [DOMEs]

FreeMoCap Foundation, Inc.

Written Proposal to the NSF X-Labs Initiative

> The mission of the proposed NSF X-Lab is to build a new class of scientific instrument — the **DOME**, which renders the complete, calibrated, synchronized interaction between an agent and its environment measurable for the first time — and the open organization that can sustain it, unifying the study of perception and action across humans, animals, and machines into a single science of real-world sensorimotor control.

Every living and engineered agent solves the same problem: it senses a thin slice of the energy available in its environment through a limited set of imperfect transducers, and on that basis generates reaction forces against an available substrate to propel itself toward its goal. Information flows in, forces flow out; the brain exists to yank the bones around. This loop — perception into action, continuously, in the real world — is the thing we have never been able to record whole.

Every revolution in science has been preceded by a revolution in what we can measure — the telescope, the microscope, and the MRI each made a new class of phenomenon knowable. Today the un-instrumented frontier is not the very small or the very distant; it is the whole behaving agent in its natural environment. The field's own stated ambition — a naturalistic, real-world science of the brain and body in action — has outrun its instruments: every thread of the perception-action loop can be measured in isolation, but none of them can yet be measured together.

Perceptuomotor neuroscience, musculoskeletal biomechanics, and legged robotics have each built superb instruments — silicon probes and miniature microscopes that resolve neural activity, markerless motion capture and force plates that resolve kinematics and kinetics, motor-unit electromyography that resolves muscle activation, mobile trackers that resolve gaze — and each measures its own thread with extraordinary precision. But no instrument captures the loop whole. The threads live in incompatible coordinate frames, on unsynchronized clocks, under semantic schemes that do not talk to each other — so the single most basic question, **what did this agent see, and what did it therefore do**, has no calibrated, unified answer. The fracture is in the measurement itself, not merely in the sociology of the fields. What is needed is a new class of instrument: one that captures every empirically available channel of an agent's perception and action at once — spatially calibrated, temporally synchronized, and rendered directly commensurable.

We call the instrument the **DOME** — a Densely Observable Measurement Environment: a densely instrumented region of real-world space engineered to record every measurable channel of the agent-environment interaction at once — full-body kinematics and kinetics, muscle and motor-unit activity, binocular gaze and reconstructed retinal input, and, where feasible, central and peripheral neural activity — all spatially calibrated, temporally synchronized, and expressed in one sensor-grounded ontology so the channels are directly commensurable rather than merely co-recorded. The DOME is one instrument in three forms. The **DOME-L** is a warehouse-scale flagship in Boston, built large enough to physically contain and cross-validate the smaller variants — so it doubles as the metrology and validation platform for the entire network rather than merely the biggest capture room. The **DOME-S** is

DRAFT - WORK IN PROGRESS

the standard, lab- and classroom-scale instrument that other groups actually deploy: the direct extension of the webcam-based capture volumes the FreeMoCap Foundation already facilitates worldwide, and the dissemination target the flagship exists to certify. The **DOME-W** is the wearable, mobile variant — an inertial suit with a mobile eye tracker and world camera — that carries calibrated capture out of the room and into natural terrain, extending the PI's outdoor gaze-and-gait lineage[1,2]. The validation path is not hypothetical: an earlier dissertation from the team already benchmarked FreeMoCap against a research-grade optical system and showed it yields clinically valid kinematics **[cite AC dissertation + preprint — keys not yet in .bib]**. The DOME-L industrializes that experiment — the flagship continuously certifies the disseminated DOME-S against itself through bootstrapping calibration, closing the loop between the instrument we build and the instrument the world runs, and letting us push a metrologically trustworthy tool to the broadest possible audience. Because the instrument is defined by its ontology rather than by any one sensor, different sensor systems hydrate the same model: a camera-based and an inertial estimate describe the same `Human` kinematics with different uncertainty and fuse into one, and the model aligns measurements across sensor generations and across species — a `Head` has `Eyes` whether or not the participant wore an eye tracker. Critically, the DOME is not a fixed rig built around a single new sensor. We advance the measurement on **both halves of the loop** the instrument records — the motor output (motion capture) and the sensory input (gaze and its retinal consequences) — so the platform's own capability frontier moves forward across the whole interaction, and the instrument itself tells us which transducer to build next. On the motion-capture side we build three things. The first is an **actuated, self-calibrating camera array**: daisy-chainable Power-over-Ethernet cameras with modular, swappable sensors and controlled infrared and visible illumination, each on a pitch-roll-yaw mount with programmatically controllable zoom, aperture, and focus — and, decisively, the entire array's extrinsics and intrinsics are controllable at once from a single console. An experimenter selects any sub-volume of the DOME in software and the whole array re-aims and re-focuses on it. Pan-tilt-zoom cameras exist and calibrated-but-fixed motion-capture cameras exist; what does not exist is a camera array that **moves and stays metrically calibrated as a coordinated measurement instrument**. The problem is real and lived: in large capture volumes, physically re-aiming and re-calibrating cameras is so labor-intensive that rigs simply get frozen in place — making the array retaskable yet always-calibrated is what lets one instrument serve both whole-room and zoomed-in measurement, and what makes it shippable to others. The second is **sensor fusion** that crosses the threshold to trustworthy inverse dynamics. Clinically valid kinematics is not the same bar as inverse dynamics — recovering joint torques and muscle forces demands accuracy that markerless capture does not yet reach, and that inertial systems only fake through drifty, black-box gap-filling. But the two modalities fail in opposite directions: an outside-in camera estimate is *accurate but noisy* (centered on truth, low precision), while an inertial estimate is *precise but drifting* (smooth, low accuracy). Fuse their complementary error profiles — an inertial suit worn inside the camera volume — and the joint estimate beats either alone, carrying explicit uncertainty rather than hidden guesswork, finally making markerless muscle-force estimation trustworthy enough for musculoskeletal modeling[3]. The third turns the instrument on the models themselves. The calibrated array gives us what the computer-vision pose-estimation field lacks at scale: physical ground truth. Using **reprojection error** — the disagreement between each camera's two-dimensional estimate and the reconstructed three-dimensional solution — as a correction signal, the DOME improves the underlying skeleton trackers everyone depends on, both as standalone trackers and inside a calibrated volume. This matters now because the field has plateaued: it drifted away from the pixels toward ungroundable targets, and its trackers, trained against aging benchmark data, still fail on out-of-distribution movement — patient populations, and extreme athletic motion such as gymnastics and circus arts. The DOME-L can record exactly those movements with ground truth, producing corpora and correction signal that measurably improve existing models — an instrument, in the Topic's own words, intentionally engineered to produce better AI training data. Further out, the same fusion makes a genuinely mobile capture conceivable — a synchronized swarm of camera drones following an inertially suited walker outdoors, its drone-borne cameras necessarily lower-fidelity than fixed walls but still metrically useful once fused with the suit and a head-mounted stereo pair. We point at this as a consequence of the core work, not a Phase-1 promise. If motion capture is the mature half we sharpen, the sensory half is where we build the flagship new transducer. Current mobile eye trackers are old, closed, and top out near a degree of error — and whole degrees of freedom, ocular torsion and lens accommodation, go **unmeasured at any price**. These are not optional refinements. Torsion is active during essentially all natural locomotion through the vestibulo-ocular reflex, and any faithful reconstruction of what actually lands on the retina requires it — the PI's own work on retinal optic flow is the paradigm case[2,4,5]. The bet is a **camera-quality bet**, **not a physics bet**: smartphones prove that tiny, fast, cheap, high-resolution imagers exist; eye trackers simply never adopted them. Put modern imaging behind controlled illumination and the two "unreachable" degrees of freedom become tractable in a mobile tracker. The payoff compounds. The ontology defines slots — torsion, accommodation — that no existing sensor fills, so the instrument itself specifies which sensor to build next; and once built, the archive improves retroactively, as models trained on the newly measurable channels back-predict them for older recordings. The instrument gets better at seeing its own past. This is track record, not aspiration: the team has already built plausibly best-in-world eye trackers for ferret and mouse, and is building the functionally equivalent human, mouse, and ferret instruments that let the same measurement cross species.

Because every stream is modeled into one ontology, DOME recordings are, with little reshaping, directly consumable by modern reinforcement-learning and robotics training stacks — a measured bone segment and a simulated robot link are the same `RigidBody`. This makes the instrument a source of grounded training data for embodied AI, and it makes the loop close: the instrument yields structured data, the data trains control models, and because a system optimized for the same task under the same constraints tends to converge on the same internal solution, those models become **testable hypotheses** about the neural computation behind the behavior **[cite Yamins/DiCarlo goal-driven modeling + motor-cortex RNN work — keys not yet in .bib]**. The hypotheses are then probed back inside the DOME by controlled perturbation of the perception-action loop itself. A modular augmented-reality ground plane — reconfigurable LED floor panels, projection, and virtual reality that reshape the terrain a real agent navigates *while it is being measured* — turns passive observation into active experiment, extending an apparatus the PI first built and published a decade ago[6–9]. Observation, modeling, and intervention become a single instrument: the full hypothetico-deductive cycle, in one apparatus.

An instrument like this cannot be built inside the institutions that most need it. A trainee's job is to *learn* by doing science; a research tool's job is to *encode* hard-won mastery so others can do science without re-learning everything — and these point in opposite directions. Academia's incentive structure runs the apprenticeship backwards: the experienced builder is rewarded for novel findings and pushed to offload infrastructure onto trainees who rotate out on degree timelines, so the shared instrument never accumulates the mastery that would make it trustworthy. The result is the well-documented graveyard of abandoned research software[10,11]. The scope of the tool is what determines whether it becomes a commons. A tool defined by a research *domain* partitions its users; a tool defined by a *measurement* unites everyone who needs that measurement, however different their goals. Scoping the X-Lab to the measurement apparatus — and deliberately refusing to own any single research question — is what lets otherwise disjoint communities meet on one instrument[12]. The FreeMoCap Foundation, founded in 2021, already ran this experiment. Scoped to the measurement itself — turning cameras and light into calibrated, usable skeleton estimates, with an obsessive focus on usability — and refusing to claim a research domain, it became a shared commons where biomechanists, neuroscientists, roboticists, animators, and game designers now meet and cross-pollinate. Domain-scoped tools such as OpenCap and DeepLabCut[13] are excellent and, precisely by being domain-scoped, are the foils that prove the point. The DOME is that proven model — measurement-scoped, master-built, open, full-time-maintained — aimed at the far larger target of the whole interaction loop. This fixes the operating model, too. The X-Lab aggressively scopes its own deliverable to the *instrument* and fans the science out to a standing network of collaborators — human perception-and-action researchers, robotics and prosthetics groups, visual neuroscientists, and animal labs spanning mouse, ferret, and guinea fowl — who use the tools we build and maintain. The community then convenes semi-annually to share findings, feed insight back into the instrument, and co-plan the next development cycle: a research culture organized around a shared, evolving apparatus rather than a stream of disconnected papers. Building the tool and leaving the science to others is not a retreat from impact; it is the mechanism of it. The instrument reshapes the field by making its measurements commensurable and cumulative, and the closed loop is the proof of what the tool makes possible — while the X-Lab's own success metric stays a working, shared instrument rather than a publication count. That metric, and the full-time career engineers held for years that it requires, is precisely the X-Labs model and precisely what a university cannot provide.

The end-state is a single empirical science of sensorimotor control. Perceptuomotor neuroscience, musculoskeletal biomechanics, mobile robotics, and embodied AI today study the same loop with incompatible tools and vocabularies; a shared instrument makes their measurements commensurable, so a finding in one becomes evidence in another — collapsing four parallel literatures into one cumulative science of real-world behavior. On the very same measurements sits the second end-state: the metrologically grounded, uncertainty-tagged corpora that embodied AI and robotics need to learn real-world sensorimotor behavior — an instrument intentionally engineered for the next generation of AI training pipelines. One apparatus, reshaping a field of science and seeding a sector of technology at once.

There'll be a Full Page Figure here

Figure 1: Blah blah blah captions. Blah blah blah captions. Blah blah blah captions. Blah blah blah captions. Blah blah blah captions.

1. Technology Landscape

- Existing tools and methods
  - Myriad single-modality hyperspecialized research threads
    - OpenSim - Widely used but universally hated by practictioners
    - DeepLabCut - Global success on paper, universally a headache in practice
    - etc
  - MuJoCo/IsaacLab Driving robotics
    - Integrate NeuroBehaioral Data into those sinulation/RL training enviornments to cross-align perceptuomotor neuroscience, HNHA biomechanics, robotics control research (e.g. Cassie from Guinea Fowl)
- New approaches to complexity management
  - Ontology/Entity Management (Anduril/Palantir/Maven Smart System)
  - Industry and Military technology applied to scientific domain
  - Want to establish an operational ontology as backbone of this tool, following powerful patterns established by (Anduril/Palantir/Maven) systems
- FreeMoCap and its landscape
  - Established Tool with Growing International Community
  - Existence is proof-of-concept of approach, XLabs scale support would accelerate FreeMoCap model into the global scale
- Laser Skeleton research
  - [1,2,4]
  - etc
  - Ferret work
    -

2. Outcomes

- Cool Tool - ~~Skinner~~SkellyBox: Multiscale modular empirical capture arena
  - Interoperabilty across species, domains, and timescales
    - A member of a mouse lab can walk into a marmoset lab and expect their pipelines will still work
    - Perceptuomotor behavior and kinematic/kinetic analysis directly interoperable with robotics models (via MuJoCo and Nvidia Isaac Gym) - Data for reinforcement learning models, Controls to interpret HNHA behavioral data
    - Data models alignment across timescales - developmental/longitudinal (months/years/lifespan) timescales aligned with perception/action timescales (seconds/minutes)
  - Composable complexity
    - Careful ontology allows for complexity via comopsible data sources and assocaited downstream entities
    - Adding new sensor modalities into established scene, framework handles spatiotemporal alignment/calibration, unit conversions, resampling, etc
- Organize and operationalize a new culture of science around discussion/management/extension shared platform tool
  - In the form of an annual Congress meeting
  - Organizing the culture of science around the discussion and advancement of shared ontologies rather than an endless stream of 8-10 page pdfs
  - Break the Skill Ceiling in academic science
    - Build tool to allow multi-decade continuous education across disciplines and experience (same tool in high school as used in federally funded research lab)
    - Shared toolset encourages natural blending of disciplines while encouraging and reward higher specialized exploration (esp when that research produces actionable results across different dimensional domains)
- Mesoscale revolution
  - Allow advances in reductionist microscale research to align with human-scale interests
  - Align Perceptuomotor Neuroscience with humanoid robotics, allow transfer, technology, and technique transfer between those fields

2.1. Timeline

Phase	Name	Description
1	**Organize**	Set up Organization (Phase 0) — Solidify release of FreeMoCap v2
2	**Architect**	Develop core software architecture, define key ontologies and entity/action pairings via use-oriented development (ferret, mouse, guinea fowl, marmoset, human research). Design and prototype instrumentation (mocap/eye cameras, electrode arrays and miniscopes, motor-unit EMG, environmental control systems)
3	**Execute**	Integrate new architecture into global/public facing community of users (presented as FreeMoCap v3). Iterative design of instrumented system
4	**Stabilize**	Establish OODA loop within expected standard operations
5	**Expand**	Once stabilized, expand key performance metrics
6	**Align**	Focus on building deeply integrated community of users and contributors (Transition BDFL ⟶ BDFaSPoT)
7	**Establish**	Focus on long term stability and healthy growth
8	**Maintain**	Beyond proposal timeline — Establish organization practice in service of unbounded infinite timeline. Apply winning model to support extending 'mesoscale research revolution' to all biological sciences (e.g. integrating deeper microbio/biochem insights beyond the scope of the present perceptuomotor/biomech/robotics focus)

Table 1: Timeline of Phased Milestones

DRAFT - Work in Progress

3. Senior/Key Personnel Qualifications

3.1. Jonathan Matthis, PhD [Principle Investigator]

3.2. EI

3.3. AC,PhD

3.4. RR, CPA or w/e

Name / Role	Qualifications
Jonathan Matthis, PhD President/CEO	• Founder and chief maintainer of FreeMoCap project • Left tenure track academic research position precisely because the institution could not support the work that needed to be done • Decades experience integrating visual neuroscience with biomechanics/robotics[1,2,4]
EI CTO	• Decades of experience in software Industry • Chief architect and CTO for established companies • Represents expertise truly absent from academic contexts • Ensures world-class enterprise-level software capability • Will build training/workshop/hackathon backend to inject EI-level software development expertise into broad community of science
AC, PhD CSO	• Expertise in validation of clinical research tools and bench-to-bedside development • [CITE DISSERTATION] • Experience in neuroprosthetics design
RR CFO	• Bookkeeping • NSF Grant management • Finances • Entrepreneurship

Table 2: Senior/Key Personnel Qualifications

4. Team Capabilities Statement

4.1. Senior Team Capabilities

- Jon - done it before in humans, doing it in ferrets and mice
- AC - focusing clinical adoption and bench-to-bedside transitions
- EI - Deep professional experience in Tech Sector, high level software architecture experienced necessary to develop truly global scale enterprise level of software in open-source/academic contexts
- RR - Organizational health and entrepreneurial development

4.2. Collaborators

Initials	Model / Domain	Expertise
BS	Ferrets	Visual/Perceptual neuroscience, ephys
MD	Guinea Fowl	Musculoskeletal Biomechanics, muscle units
DF	Mice	System Biology
AH	Marmosets / NHP	Ephys
MH	Human	Vision and CPS
Mayo Clinic	Human	Brain scans and implanted sensors
BD, GN/RAI, CH, JH	Robots / Fabrication	Control theory
AA	Physical facilities	Rapid iteration, recruiting

Table 3: Collaborator Network

4.3. Governance

- Most successful FOSS project operate under a Benevolent Dictator For Life (BDFL) Governance (e.g. Linus Torvald at Linux)
- This is best when an organization is young, but later on centralization is a concern control should melt into the community of users (see Ton Roosenthaal's step down from Blender lead position)
- We will use Benevolent Dictator for a Set Period of Time (BDfaSPoT)
  - Transition from centralized control: President -> President + Board -> to President + Board (C-Suite) + Community DAO
    - President maintains at least 51% control until year 5, then optional transition to 50/50 President/Board+Community
- Plan changes via public Requests for Comments (similar to Python PEP system)
  - Provide public space for feedback on planned features, allows transparency without creating a bottleneck
- In 2nd year, create Developer Fund and Community Grants program
  - Stakeholders vote on proposals and features, via DAO like voting
- In 4-5th year, extend this system to handle community steering of org

4.4. Autonomy

- We are already autonomous, as of 2 weeks prior to the deadline of this proposal.
  - This project will happen no matter what, but with XLabs funding, we could truly change the world

Bibliography

1. Matthis, J. S., Yates, J. L. & Hayhoe, M. M. Gaze and the Control of Foot Placement When Walking in Natural Terrain. *Current Biology* **28,** 1224–1233.e5 (2018).
2. Matthis, J. S., Muller, K. S., Bonnen, K. L. & Hayhoe, M. M. Retinal Optic Flow during Natural Locomotion. *PLOS Computational Biology* **18,** e1009575 (2022).
3. Delp, S. L. *et al.* OpenSim: Open-Source Software to Create and Analyze Dynamic Simulations of Movement. *IEEE Transactions on Biomedical Engineering* **54,** 1940–1950 (2007).
4. Muller, K. S. *et al.* Retinal Motion Statistics during Natural Locomotion. *eLife* **12,** e82410 (2023).
5. Muller, K. S. *et al.* Foothold Selection during Locomotion in Uneven Terrain: Results from the Integration of Eye Tracking, Motion Capture, and Photogrammetry. https://elifesciences.org/reviewed-preprints/91243v2 (2024) doi:10.7554/eLife.91243.2.
6. Matthis, J. S. & Fajen, B. R. Humans Exploit the Biomechanics of Bipedal Gait during Visually Guided Walking over Complex Terrain. *Proceedings of the Royal Society B: Biological Sciences* **280,** 20130700 (2013).
7. Matthis, J. S. & Fajen, B. R. Visual Control of Foot Placement When Walking over Complex Terrain. *Journal of Experimental Psychology: Human Perception and Performance* **40,** 106–115 (2014).
8. Matthis, J. S., Barton, S. L. & Fajen, B. R. The Biomechanics of Walking Shape the Use of Visual Information during Locomotion over Complex Terrain. *Journal of Vision* **15,** 10 (2015).
9. Matthis, J. S., Barton, S. L. & Fajen, B. R. The Critical Phase for Visual Control of Human Walking over Complex Terrain. *Proceedings of the National Academy of Sciences* **114,** E6720–E6729 (2017).
10. Eghbal, N. *Working in Public: The Making and Maintenance of Open Source Software.* (Stripe Press, 2020).
11. Barker, M., Katz, D. S., Gonzalez-Beltran, A. & others. Better Incentives Are Needed to Reward Academic Software Development. *Nature Ecology & Evolution* **7,** 626–627 (2023).
12. Star, S. L. & Griesemer, J. R. Institutional Ecology, `Translations' and Boundary Objects. *Social Studies of Science* **19,** 387–420 (1989).
13. Mathis, A. *et al.* DeepLabCut: Markerless Pose Estimation of User-Defined Body Parts with Deep Learning. *Nature Neuroscience* **21,** 1281–1289 (2018).